

MISC

Quadratic formula:

$$ax^2 + bx + c = 0 \Rightarrow x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Monotonically in-/decreasing:

$$\forall x. (f'(x) > 0) \Rightarrow a > b \Leftrightarrow f(a) > f(b)$$
$$\forall x. (f'(x) < 0) \Rightarrow a > b \Leftrightarrow f(a) < f(b)$$

RSS:

$$RSS = \sum_i (y_i - f(x_i))^2$$

TRIGONOMETRY

x	0	$30^\circ = \pi/6$	$45^\circ = \pi/4$	$60^\circ = \pi/3$	$90^\circ = \pi/2$
$\sin(x)$	0	$1/2$	$\sqrt{2}/2$	$\sqrt{3}/2$	1
$\cos(x)$	1	$\sqrt{3}/2$	$\sqrt{2}/2$	$1/2$	0
$\tan(x)$	0	$1/\sqrt{3}$	1	$\sqrt{3}$	

$\tan(x) = \frac{\sin(x)}{\cos(x)}$
$$\sin(2x) = 2 \sin(x) \cos(x)$$
$$\sin^2(x) + \cos^2(x) = 1$$
1, sin(α), cos(α)

VECTORS

Term	Definition
Norm/Length	$\ v\ = \sqrt{\sum_{i=1}^n v_i^2} = \sqrt{v^\top v}$ $\nabla \ v\ = \frac{1}{\ v\ } v^\top$ $\nabla \frac{1}{\ v\ } = -\frac{1}{\ v\ ^2} \nabla \ v\ = -\frac{1}{\ v\ ^3} v^\top$ $\nabla \ r\ ^2 = 2[r] \frac{1}{\ r\ } r^\top = 2r^\top$
Distance	$d(u, v) = \ u - v\ $ $\nabla d(u, v) = \frac{1}{d(u, v)} ((u - v)^\top (v - u)^\top)$
Scalar product	$u \bullet v = u^\top v = \sum_k u_k v_k$ $\nabla (u \bullet v) = (v^\top u^\top)$

MATRICES

Affine transformation:

$$M \in \mathbb{R}^{m \times n}, \quad A_{b,M} : \begin{cases} \mathbb{R}^m \rightarrow \mathbb{R}^n \\ x \mapsto b + M \cdot x \end{cases}$$

Determinant:

$$\det(a) = a$$
$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - cb$$
$$\det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}$$
$$\det(A \cdot B) = \det(A) \cdot \det(B) \quad \det(\mathbb{1}) = 1$$
$$\det(A^{-1}) = \frac{1}{\det(A)} \quad \det(A^\top) = \det(A)$$

Inverting:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow A^{-1} = \frac{1}{ad - cb} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Transposing:

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 1 \end{pmatrix}^\top = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 2 & 1 \end{pmatrix} \quad (A^\top)^\top = A \quad (A+B)^\top = A^\top + B^\top$$
$$\begin{pmatrix} 1 \\ 4 \\ 5 \end{pmatrix}^\top = (1 \ 4 \ 5) \quad (\lambda A)^\top = \lambda A^\top \quad (AB)^\top = B^\top A^\top$$

Matrix multiplication:

$$\begin{pmatrix} 2 & 3 & 1 \\ 4 & -1 & 7 \end{pmatrix} \cdot \begin{pmatrix} 6 & 4 \\ 1 & 0 \\ 8 & 9 \end{pmatrix} = \begin{pmatrix} (2 \cdot 6 + 3 \cdot 1 + 1 \cdot 8) & (2 \cdot 4 + 3 \cdot 0 + 1 \cdot 9) \\ (4 \cdot 6 + -1 \cdot 1 + 7 \cdot 8) & (4 \cdot 4 + -1 \cdot 0 + 7 \cdot 9) \end{pmatrix} = \begin{pmatrix} 23 & 17 \\ 79 & 79 \end{pmatrix}$$

PROPERTIES

$$A \cdot A^{-1} = \mathbb{1} \quad \ker A = \{v \mid Av = 0\}$$
$$A + B = B + A \quad A \cdot B \neq B \cdot A$$
$$A + (B + C) = (A + B) + C \quad A(BC) = (AB)C$$
$$A(B + C) = AB + AC \quad (A + B)C = AC + BC$$

INDICES

$$A \in \mathbb{R}^{n \times m}, B \in \mathbb{R}^{m \times r}, \quad A \cdot B \in \mathbb{R}^{n \times r}$$
$$(A \cdot B)_{ij} = \sum_{k=1}^m A_{ik} B_{kj} = \sum_k A_{ik} B_{kj}$$
$$((A \cdot B)^\top)_{ij} = (A \cdot B)_{ji} = \sum_k A_{jk} B_{ki}$$
$$(A \cdot B^\top)_{ij} = \sum_k A_{ik} B_{kj}^\top = \sum_k A_{ik} B_{jk}$$
$$a^\top \cdot B \cdot b = \sum_{ij} (a^\top)_i B_{ij} b_j = \sum_{ij} a_i B_{ij} b_j$$
$$(B \cdot a)(C \cdot a) = \sum_i (B \cdot a)_i (C \cdot a)_i$$

ADDITION/SUBTRACTION

$$A \in \mathbb{R}^{n \times m}, B \in \mathbb{R}^{n \times m}$$
$$(A + B)_{ij} = A_{ij} + B_{ij}$$
$$(A - B)_{ij} = A_{ij} - B_{ij}$$

KRONECKER DELTA

$$\delta_{ij} = (\vec{e}_i)_j = \begin{cases} 1, & i = j \\ 0, & \text{otherwise} \end{cases}$$
$$(E_{ij})_{kr} = \delta_{ik} \delta_{jr}$$
$$(E_{rst})_{ijk} = \delta_{ri} \delta_{sj} \delta_{tk}$$

BASIS VECTORS

$$\mathbb{R}^3$$
$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

UNIT MATRICES

$$\mathbb{R}^2$$
$$E_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, E_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, E_{21} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, E_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

AFFINE TRANSFORMATIONS

$$A_{a,M}(x) = a + M \cdot x$$
$$L_A(\vec{0}) = \vec{0}$$

DERIVATIVES

1	→ 0	x	→ 1
x ²	→ 2x	x ⁿ	→ nx ⁿ⁻¹
1/x	→ -1/x ²	√x	→ 1/(2√x)
e ^x	→ e ^x	e ^{-x}	→ -e ^{-x}
ln(x)	→ 1/x, x > 0	ln(y · x)	→ 1/x, x > 0
a ^x	→ ln(a) · a ^x , a > 0, a ≠ 1	log _b (x)	→ 1/ln(b) · x
sin(x)	→ cos(x)	sin(2x)	→ 2 cos(2x)
cos(x)	→ -sin(x)	cos(ax)	→ -a sin(ax)
tan(x)	→ 1 + tan ² (x) = 1/cos ² (x)	arccot(x)	→ -1/(1+x ²)
cot(x)	→ -1/sin ² (x)	arcsin(x)	→ 1/√(1-x ²)
arccos(x)	→ -1/√(1-x ²)	arctan(x)	→ 1/(1+x ²)

Case	Rule
Addition	$(f(x) + g(x))' = f'(x) + g'(x)$
Multiplication	$(\alpha \cdot f(x))' = \alpha \cdot f'(x)$
Product rule	$(f \cdot g)' = f' \cdot g + f \cdot g'$ $(f \cdot g \cdot h)' = f' \cdot g \cdot h + f \cdot g' \cdot h + f \cdot g \cdot h'$
Chain rule	$(f(g(x)))' = f'(g(x)) \cdot g'(x)$
Quotient rule	$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$

FUNCTIONS OF SEVERAL VARIABLES

Term	Definition
Real valued function	$R \subset \mathbb{R}$
Vector valued function	$R \subset \mathbb{R}^m$
A function of n variables	$D \subset \mathbb{R}^n$
n-dimensional curve	$c: \mathbb{R} \rightarrow \mathbb{R}^n$
n-dimensional hyper-surface	$s: \mathbb{R}^n \rightarrow \mathbb{R}$
Reparametrization	$c(t) \rightarrow d(s) = c(h(s))$
Softmax function	$\sigma: \left\{ \mathbb{R}^n \rightarrow [0; 1]^n \right. \\ \left. x \mapsto e^x, \left(\sum_{j=1}^n e^{x_j} \right)^{-1} \right\}$

Sigmoid function:

$$\sigma: \begin{cases} \mathbb{R} \rightarrow [0; 1] \\ x \mapsto \frac{1}{1+e^{-x}} = \frac{e^x}{1+e^x} \end{cases}$$
$$\sigma(-x) = 1 - \sigma(x)$$
$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$

DERIVATIVES

Function	Derivative
$f: \begin{cases} \mathbb{R} \rightarrow \mathbb{R}^n \\ x \mapsto y \end{cases}$	$f': \begin{cases} \mathbb{R} \rightarrow \mathbb{R}^n \\ x \mapsto y \end{cases} = \begin{pmatrix} \frac{d}{dx} f_1(x) \\ \frac{d}{dx} f_2(x) \\ \vdots \\ \frac{d}{dx} f_n(x) \end{pmatrix}$
$f: \begin{cases} \mathbb{R}^n \rightarrow \mathbb{R} \\ x \mapsto y \end{cases}$	$f': \begin{cases} \mathbb{R}^n \rightarrow \mathbb{R}^{1 \times n} \\ x \mapsto y \end{cases} = \nabla f$ $= \left(\frac{\partial}{\partial x_1} f(x), \frac{\partial}{\partial x_2} f(x), \dots, \frac{\partial}{\partial x_n} f(x) \right)$
$f: \begin{cases} \mathbb{R}^n \rightarrow \mathbb{R}^m \\ x \mapsto y \end{cases}$	$f': \begin{cases} \mathbb{R}^n \rightarrow \mathbb{R}^{m \times n} \\ x \mapsto y \end{cases} = J_f = \frac{\partial f}{\partial x} \Big _{x=x_0}$ $= \begin{pmatrix} \nabla f_1(x) \\ \vdots \\ \nabla f_m(x) \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial x_1} f_1(x) & \dots & \frac{\partial}{\partial x_n} f_1(x) \\ \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1} f_m(x) & \dots & \frac{\partial}{\partial x_n} f_m(x) \end{pmatrix}$

PARTIAL VS TOTAL DERIVATIVE

$$f(x, y) = x^2 + 2y$$
$$\frac{\partial f}{\partial x} = 2x$$
$$\frac{df}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dx} = 2x + 2 \cdot \frac{dy}{dx} \quad (y \text{ dependent on } x)$$

GRADIENT VS TOTAL DERIVATIVE

$$f(x, y) = xye^y$$
$$x(t) = e^t$$
$$y(t) = t^2$$
$$\nabla f = (ye^y, xe^y + xye^y) = e^y(y, x + xy)$$
$$\frac{d}{dt} f = e^y(y, x + xy) \cdot \frac{d}{dt} \begin{pmatrix} e^t \\ t^2 \end{pmatrix} = \frac{d}{dt} (e^t t^2 e^t) = (2t + t^2 + 2t^3) e^{t^2+t}$$

GRADIENTS

Common gradients:

$$f: \mathbb{R}^n \rightarrow \mathbb{R}, x \in \mathbb{R}^n, \quad a \in \mathbb{R}^n, c \in \mathbb{R}, A \in \mathbb{R}^{n \times n}$$
$$f(x) = \|x\|^2 \Rightarrow \nabla f = 2x^\top$$
$$f(x) = a^\top x + c \Rightarrow \nabla f = a^\top$$
$$f(x) = x^\top A x \Rightarrow \nabla f = x^\top (A + A^\top)$$

$$f(x) = \frac{1}{2} x^\top A x + b^\top x + c \Rightarrow \nabla f = x^\top \frac{A + A^\top}{2} + b^\top$$
$$= x^\top A + b^\top \text{ if } A^\top = A$$
$$g: \mathbb{R}^m \rightarrow \mathbb{R}, B \in \mathbb{R}^{m \times n}$$
$$f(x) = g(Bx + b) \Rightarrow \nabla f = (\nabla g(Bx + b))^\top B$$

JACOBIAN MATRIX

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad x_0 \in \mathbb{R}^n$$
$$J_f(x_0): \mathbb{R}^{m \times n}, \quad J_f(x_0)_{ij} = \frac{\partial}{\partial x_j} f_i(x_0)$$
$$f: \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad J_f(x, y) = \begin{pmatrix} \frac{\partial x_1}{\partial x_2} & \frac{\partial y_1}{\partial y_2} \end{pmatrix}$$

ty lukas <3

$$\begin{array}{ccc} \text{Euclid}_* & & \text{Mat}_{\mathbb{R}} \\ \begin{array}{c} (\mathbb{R}^n, x_0) \\ \downarrow g \\ (\mathbb{R}^m, g(x_0)) \\ \downarrow f \\ (\mathbb{R}^k, f(g(x_0))) \end{array} & \xrightarrow{f \circ g} & \begin{array}{c} \mathbb{R}^n \\ \downarrow J_g(x_0) \\ \mathbb{R}^m \\ \downarrow J_f(g(x_0)) \\ \mathbb{R}^k \end{array} \end{array}$$

Chain rule:

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m, g: \mathbb{R}^m \rightarrow \mathbb{R}^k, h: \mathbb{R}^n \rightarrow \mathbb{R}^k = g \circ f$$
$$J_h = J_g(f(x)) \cdot J_f(x)$$

Common jacobians:

$$f(x) = a + M \cdot x \Rightarrow J_f = M$$
$$f(x) = x \Rightarrow J_f = \mathbb{1}$$
$$f(x) = \|x\|^2 \Rightarrow J_f = (2x)^\top$$

LINEARISATION

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m, x_0 \in \mathbb{R}^n$$
$$L_f(x) = f(x_0) + J_f(x_0) \cdot (x - x_0)$$

For finding better approximations for x₀:

1) Calculate J_f, then J_f(x₀), then L_f

2) Find x so that L_f(x) = 0 (Gauss)

For calculating an approximate value at x:

1) x₀ = round x to ints

2) Calculate L_f(x) at x₀

SECOND DERIVATIVE

$$\frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} (f(x, y)) \right) = \frac{\partial^2 f(x, y)}{\partial x^2} = \partial_{xx} f(x, y)$$

HESSIAN MATRIX

$$f: \mathbb{R}^n \rightarrow \mathbb{R}, \quad x \in \mathbb{R}^n, \quad H(x) \in \mathbb{R}^{n \times n}$$
$$H(x, y) = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} = J(\nabla f)$$
$$(H(x))_{ij} = \frac{\partial}{\partial x_i} \frac{\partial}{\partial x_j} f(x)$$
$$\partial_i \partial_j f = \partial_j \partial_i f$$

Common Hessians:

$$f(x) = \frac{1}{2} x^\top A x + b^\top x + c \Rightarrow H_f = \frac{A + A^\top}{2}$$
$$f(x) = g(Ax + b) \Rightarrow H_f = A^\top H_g(Ax + b) A$$
$$g: \mathbb{R}^m \rightarrow \mathbb{R}$$

EXTREMAL POINTS

Quadratic form:

$$M \in \mathbb{R}^{n \times n}, \quad q_M(v) = v^\top M v \quad v \in \mathbb{R}^n$$
$$H = \frac{1}{2} (M + M^\top) \quad \forall v \in \mathbb{R}^n. (q_M(v) = q_H(v))$$

$v^\top H v > 0 \Rightarrow c_0$ is a **local minimum**, H is **positive definite**

$v^\top H v \geq 0 \Rightarrow c_0$ is a **local minimum** or **non-extremal**, H is **positive semi-definite**

$v^\top H v < 0 \Rightarrow c_0$ is a **local maximum**, H is **negative definite**

$v^\top H v \leq 0 \Rightarrow c_0$ is a **local maximum** or **non-extremal**, H is **negative semi-definite**

$v^\top H v \neq 0 \Rightarrow c_0$ is a **saddle point**, H is **indefinite**

Finding stationary points:

1) Calculate ∇f, then H_f

2) Find stationary point(s) x so that ∇f(x) = 0

3) Find type of point using technique of completing squares on H_f(x)

$$(v_1, v_2, v_3) \cdot \begin{pmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{pmatrix} \cdot \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$
$$= v_1^2 h_{11} + v_2^2 h_{22} + v_3^2 h_{33} + 2v_1 v_2 h_{12} + 2v_2 v_3 h_{23} + 2v_1 v_3 h_{13}$$

Technique of completing squares:

$$\frac{x^2}{\sqrt{c}} - \frac{4xy}{\sqrt{c}} + 1$$
$$= (x - 2y)^2 - (-2y)^2 + 1$$
$$= (x - 2y)^2 - 4y^2 + 1$$

HYPER-SURFACES

Chain rule:

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m, g: \mathbb{R}^m \rightarrow \mathbb{R}, h: \mathbb{R}^n \rightarrow \mathbb{R} = g \circ f$$
$$\nabla h(x) = \nabla g(f(x)) = \nabla g(f) |_{f=f(x)} \cdot \frac{\partial}{\partial x} f(x)$$

Differentiation properties:

$$f: \mathbb{R}^n \rightarrow \mathbb{R}, g: \mathbb{R}^n \rightarrow \mathbb{R}$$
$$\nabla(f + g) = \nabla f + \nabla g$$
$$\nabla(f - g) = \nabla f - \nabla g$$
$$\nabla(f \cdot g) = g \nabla f + f \nabla g$$
$$\nabla \frac{f}{g} = \frac{g \nabla f - f \nabla g}{g^2}$$

COMMUTATIVE DIAGRAMS

$$f: D \rightarrow M$$
$$g: N \rightarrow R$$
$$h: D \rightarrow R$$
$$= g \circ f$$

GRADIENT DESCENT / NEWTON'S METHOD

$$GD \quad x_{i+1} = x_i - \gamma \cdot \nabla f(x_i)$$
$$N \quad x_{i+1} = x_i - \gamma \cdot (H_{f(x_i)})^{-1} \cdot \nabla f(x_i)$$

γ = step size

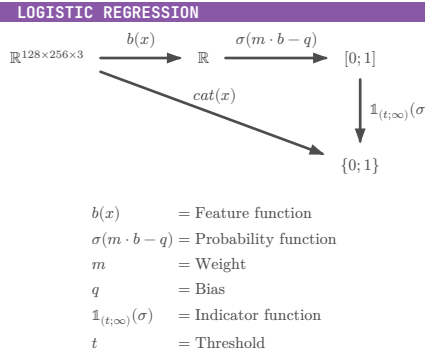
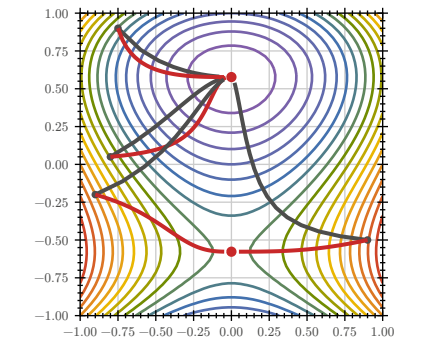
termination criterion ε > |x_{i+1} - x_i|

Gradient descent doesn't always produce global minimum:

– If the function has more than one local minimum it could output that → run with different starting points

– Could output a stationary point that is not minimal (unlikely). → run with different starting points

– Could not terminate at all if step size / learning rate is so big (overshoots minimal point). Likely to happen if the graph of the function is narrow around the minimal point. → choose smaller step size



- TODO:**
- FS2023 4)
 - likelihood function
 - maximum likelihood principle
 - find θ from $x_1, \dots, x_n$ from $F(x|\theta)$
 - $\frac{d}{d\theta} - \ln(l(\mu, \sigma)) = 0$
 - cost function $\rightarrow$ derivative $\rightarrow 0$
 - $l(p) = (\prod p) \cdot (\prod (1 - p))$

PROBABILITY THEORY	
Term	Definition
Distributed	$\sim$
Normal distribution	$\mathcal{N}(\mu, \sigma)$
Uniform distribution	$\text{unif}(a, b)$
Probability measure	$\mathbb{P}(E)$
Density	$f(x_i \mid \mu, \sigma)$
Likelihood function	$l(\mu, \sigma) = \prod_{i=1}^n f(x_i \mid \mu, \sigma)$
Log-likelihood function	$\ln(l(\mu, \sigma)) = \sum_{i=1}^n \ln(f(x_i \mid \mu, \sigma))$
Cost function (negative log-likelihood)	$-\ln(l(\mu, \sigma)) = -\sum_{i=1}^n \ln(f(x_i \mid \mu, \sigma))$
Indicator function	$\mathbb{1}_{(a;b)}(x) = \begin{cases} 1 & x \in (a;b) \\ 0 & x \notin (a;b) \end{cases}$
CDF	Cumulative Distribution Function F
PDF	Probability Density Function f

$|f_Y(y) dy| = |f_X(x) dx|$ if monotonically in/decreasing

KOLMOGOROV AXIOMS

$\mathbb{P}(\Omega) = 1$
 $\mathbb{P}(\emptyset) = 0$

$\forall A, B \subset \Omega, A \cap B = \emptyset. (\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B))$
 $\forall A, B \subset \Omega. (\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B))$

STATISTICALLY INDEPENDENT
 $X : \Omega \rightarrow \mathbb{R}$ and $Y : \Omega \rightarrow \mathbb{R}$ are **statistically independent**, if the probability density f_Y of the random variable

satisfies

$f_Y(x, y) = f_X(x) \cdot f_Y(y)$

DISCRETE PROBABILITY DISTRIBUTION
Let $\Omega = \{e_1, e_2, \dots, e_n\}$ be enumerable

$\mathbb{P}(\{e_k\}) = p_k, \quad k \in \{1, 2, \dots, n\}, \quad E \subset \Omega$

$\mathbb{P}(E) = \sum_{e \in E} p(e)$

$F(x) = \sum_{e \in \Omega, e \leq x} p(e)$

RULES

$\forall e \in \Omega. (0 \leq p(e) \leq 1)$

$\sum_{e \in \Omega} p(e) = 1$

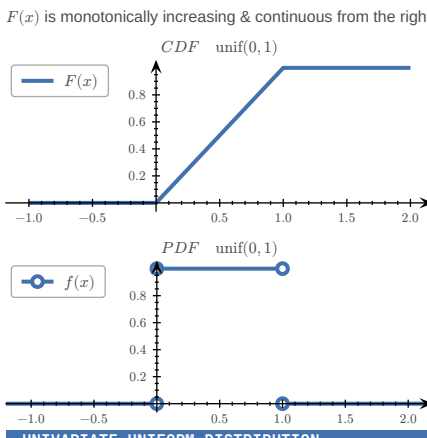
$\lim_{x \rightarrow -\infty} F(x) = 0$

$0 \leq F(x) \leq 1$

$\int_{-\infty}^{\infty} f(t) dt = 1$

$\lim_{x \rightarrow \infty} F(x) = 1$

$f(t) \geq 0$ for $t \in \mathbb{R}$



UNIVARIATE UNIFORM DISTRIBUTION

$X \sim \text{unif}(a, b), \quad \Omega \subset \mathbb{R}$

$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{if } x \in (a; b) \\ 0 & \text{else} \end{cases} = \mathbb{1}_{(a;b)}(x) \cdot \frac{1}{b-a} = F'_X$

$F_X(x) = \mathbb{P}((-\infty; x] \cap \Omega) = \mathbb{1}_{[a;b]}(x) \cdot \frac{x-a}{b-a} + \mathbb{1}_{(b;\infty)}(x)$

$= \mathbb{P}(X \leq x) = \int_{-\infty}^x f_X(t) dt$

UNIVARIATE NORMAL DISTRIBUTION

Standard normal distribution:

$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$

General normal distribution:

$f(x|\mu, \sigma) = \frac{1}{\sigma} \varphi\left(\frac{x-\mu}{\sigma}\right)$

Univariate normal distribution:

$x \sim \mathcal{N}(\mu, \sigma)$

$f(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$

$\mu \approx \bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$

$\sigma^2 \approx \text{var} = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2$

Error function:

$V : \begin{cases} \Omega \rightarrow \mathbb{R}^2 \\ \omega \mapsto \begin{pmatrix} X(\omega) \\ Y(\omega) \end{pmatrix} \end{cases}$

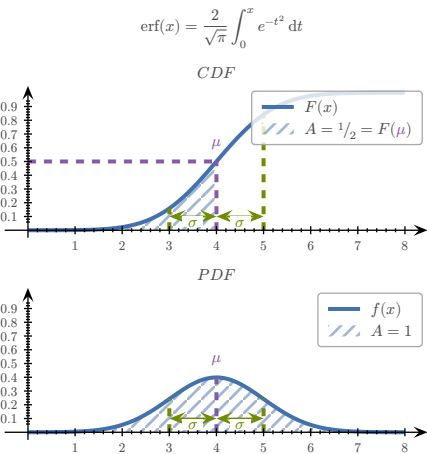
$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$

CDF

The graph shows the CDF of a normal distribution. The x-axis ranges from 1 to 8, and the y-axis ranges from 0.1 to 0.9. A blue curve represents $F(x)$. A vertical dashed line at μ intersects the curve at $A = 1/2 = F(\mu)$. The area under the curve to the left of μ is shaded blue, and the area to the right is shaded light blue.

PDF

The graph shows the PDF of a normal distribution. The x-axis ranges from 1 to 8, and the y-axis ranges from 0.1 to 0.9. A blue curve represents $f(x)$. A vertical dashed line at μ intersects the curve at $A = 1$. The area under the curve to the left of μ is shaded blue, and the area to the right is shaded light blue.



Sample mean:

$\mu \approx \bar{v} = \frac{1}{N} \sum_{i=1}^N v_i$

Sample covariance matrix:

$\Sigma \approx Q = \frac{1}{N-1} \sum_{i=1}^N (v_i - \bar{v})(v_i - \bar{v})^T$

EXAMPLES

WORKING WITH INDICES

$x, y \in \mathbb{R}^n, f : \mathbb{R}^n \rightarrow \mathbb{R}$

$f(x) = (x - y)^T (3x - y) = \sum_{i=1}^n (x_i - y_i)(3x_i - y_i)$

$\partial x_i f(x) = (3x_i - y_i) + 3(x_i - y_i)$

$f(x) = \ln(x_1 \cdot x_2 \cdot \dots \cdot x_n) - |x|^2 = \sum_{i=1}^n (\ln(x_i) - x_i^2)$

$\partial x_i f(x) = \frac{1}{x_i} - 2x_i$

$f(x) = \ln(y^T x) = \ln\left(\sum_{i=1}^n y_i x_i\right) \Rightarrow \nabla f = \frac{1}{y^T x} y^T$

CALCULATING LIKELIHOOD FUNCTION

Given the following data and the probability density of the form

x	y
1	7
2	16
3	32
4	44

find a formula for the likelihood function:

$l(\lambda) = f(7) \cdot f(16) \cdot f(32) \cdot f(44) = \lambda^4 e^{-99\lambda}$

Find a formula for the negative log-likelihood:

$-\ln(l(\lambda)) = -\ln(\lambda^4 e^{-99\lambda}) = 99\lambda - 4 \ln(\lambda)$

CALCULATING PROBABILITY DENSITY FUNCTION

EXPONENTIAL DISTRIBUTION

Given σ , Y and X , an exponentially distributed random var

$f_X(x) = X_{[0;\infty)}(x) \cdot e^{-x}$

$Y = \sigma(X)$

calculate the probability density function $f_Y(y)$

$|f_Y(y) dy| = |f_X(x) dx|$

$\frac{dy}{dx} > 0 \Leftrightarrow f_Y(y) = f_X(x) \cdot \frac{dx}{dy}$

$y = \frac{1}{1 + e^{-x}}$

$\Leftrightarrow x = -\ln\left(\frac{1}{y} - 1\right), \quad 0 < y < 1$

$\Rightarrow \frac{dx}{dy} = -\frac{1}{\frac{1}{y} - 1} \cdot \frac{-1}{y^2} = \frac{1}{y - y^2}$

$\Rightarrow f_Y(y) = X_{[0;\infty)}(x) \cdot e^{-x} \cdot \frac{1}{y - y^2}$

$= \frac{1}{y^2} \cdot X_{[0;\infty)}\left(-\ln\left(\frac{1}{y} - 1\right)\right)$

$-\ln\left(\frac{1}{y} - 1\right) \geq 0$

$y > 0 \Leftrightarrow y \geq \frac{1}{2}$

$\Rightarrow X_{[0;\infty)}\left(-\ln\left(\frac{1}{y} - 1\right)\right) = X_{[\frac{1}{2}; 1)}(y)$

$\Rightarrow f_Y(y) = \frac{1}{y^2} X_{[\frac{1}{2}; 1)}(y)$

UNIFORM DISTRIBUTION

Given $X \sim \text{unif}(0, 1), p > 0, Y = X^p$ calculate the probability density function $f_Y(y)$

$F_X(x) = \mathbb{P}(X \leq x)$

$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(X^p \leq y) = \mathbb{P}\left(X \leq y^{\frac{1}{p}}\right) = F_X\left(y^{\frac{1}{p}}\right)$

$= \mathbb{1}_{[0;1]}(y) y^{\frac{1}{p}} + \mathbb{1}_{(1;\infty)}(y)$

$f_Y(y) = \frac{d}{dy} F_Y(y) = \mathbb{1}_{[0;1]}(y) \frac{1}{p} y^{\frac{1}{p}-1}$

UNIFORM DISTRIBUTION 2

$X \sim \text{unif}(1, 3), \quad Y = \sqrt[3]{5 - X}$

$f_X(x) = \mathbb{1}_{(1;3)}(x) \frac{1}{2}$

$X = 5 - 2Y^3$

$g(y) = 5 - 2y^3$

$f_Y(y) = -f_X(g(y)) g'(y)$ monot. decr.

$= 3y^2 \mathbb{1}_{(1;3)}(5 - 2y^3)$

$1 < 5 - 2y^3 < 3$

$\Leftrightarrow 1 < y < \sqrt[3]{2}$

$\Rightarrow f_Y(y) = 3y^2 \mathbb{1}_{(1; \sqrt[3]{2})}(y)$

TODO:

- check if function is increasing ($f' > 0$) $\rightarrow$ Aufgabe 123/124
- distribution to density and vice-versa $X = g(Y) \wedge g$ increasing $\Rightarrow f_Y(y) = f_X(g(y)) \cdot g'(y)$

$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i$

$\hat{\sigma} = \left(\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2\right)^{\frac{1}{2}}$

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